

Portable bullet resistant wall

DARPA has started a new project where they aim to create a bullet resistant wall right from a portable cylinder <http://dgit.in/1IVJeJs>

**Doctors using Big Data**

Doctors are now looking into big data to predict possible health issues in the future. Read the whole article here <http://dgit.in/1IVJlOl>

To the Moon and back

Researchers are working on bringing high-speed Internet to the Moon. Read this and more stories in this month's Web Watch

Shooting lasers at the Moon for high-speed Internet

If the unavailability of Internet was the only thing keeping you from visiting the Moon, then here's an important update for you: Researchers from the Massachusetts Institute of Technology's (MIT) Lincoln Laboratory in co-ordination with NASA will demonstrate a new data communication technology that provides broadband-like connectivity for transmitting data to and from earth.

The technology will be demoed at the Conference on Lasers and Electro-Optics, scheduled to be held in July in California, US. The team will provide details and an overview of the on-orbit performance of their laser-based communication uplink between the Moon and Earth, which has reportedly surpassed the previous record transmission speed by 4,800 times.

"This will be the first time that we present both, the implementation overview and how well it actually worked," said Mark Stevens of MIT Lincoln Laboratory. "The on-orbit performance was excellent and close to what we'd predicted, giving us confidence that we have a good understanding of the underlying physics," Stevens added.

The MIT team had last year shown that their Lunar Laser Communication Demonstration (LLCD) transmitted data between the Moon and Earth at a download rate of 622 Mbps per second – said to be faster than

any radio frequency system. The team also managed to transmit data from Earth to the Moon at 19.44 Mbps per second, 4,800 times quicker than the best RF uplink ever used.

"LLCD also demonstrated the ability to 'hand-off' the laser connection from one ground station to another, just as a cellphone does a hand-off from one cell tower to another. An additional achievement was the ability to operate LLCD without using LADEE's radio at all," read a statement from NASA last year.

"Communicating at high data rates from Earth to the Moon with laser beams is challenging because of the 400,000km distance spreading out the light beam," Stevens said. "It's doubly



Data can be sent between the Moon and Earth at a download rate of 622 Mbps per sec

difficult going through the atmosphere, because turbulence can bend light – causing rapid fading or dropouts of the signal at the receiver," said Stevens.

To ensure smooth signal over such a massive distance, the team used several techniques to accomplish error-free performance over several optical

challenging conditions in both, dark and bright conditions. A ground terminal, set up at New Mexico, uses four separate telescopes to send the uplink signal to the Moon. Each telescope is about 6 inches in diameter and fed by a laser transmitter that sends information coded as pulses of invisible infrared light.

Twitter now allows animated GIFs

Twitter now supports animated GIFs on its desktop and mobile (Android and iOS) platforms. The GIFs won't play automatically and users must press a play button accompanied by the animated images or just click on the tweet to view them.

The company made the announcement via a tweet to demonstrate the animated GIFs: "Starting today, you can share and view animated GIFs," Twitter wrote in a June tweet, which was accompanied by a GIF showing a man holding a cute fluffy dog with the caption "GIFs got me like."

Twitter is also rolling out support for its API that will allow third-party developers to offer similar support in their client applications. Several other social networks and blogging sites like Pinterest and Tumblr already offer support for animated GIFs. However, social network giants Twitter and Facebook have been the exceptions. Facebook has decided not to allow GIFs on its site because it reportedly believes that they'll make the News Feeds very chaotic.

Twitter recently introduced a feature that allows users to embed tweets within new tweets. The new feature will make tweets appear more streamlined. The functionality has been rolled out only for Android and iOS devices



The GIFs won't play automatically

for now and isn't available for desktop users. The micro-blogging site has made no comment on whether the feature will be launched on more platforms.

Twitter has also launched hubs for tracking the FIFA World Cup 2014. Then there's a new sign-up form that gives users the ability to choose their team's flag as their profile image. Users will also get access to scoreboards for each game as well as custom timelines for each individual match.

Google partners with Twitter for public disaster alerts

Google's Public Alerts service will now incorporate tweets from disaster-struck locations. If a public disaster alert from sources such as the National Weather Service is issued, it will include relevant tweets in